Symmetry Reduction in Semidefinite Optimization

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(joint work with Tillman Weisser)

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Semidefinite program

$$
\begin{aligned}
\min_{Q \in \mathcal{S}^N} \quad & \langle C, Q \rangle \\
\text{subject to:} \quad & \langle A_i, Q \rangle = b_i, \quad i = 1, 2, \ldots, m \\
& Q \succeq 0.
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Is $p(x) = x^6 - 2x^4 + 2x^2$ always non-negative? $\rightarrow$ sum of squares relaxation: Does there exist psd matrix Q s.t.

$$p(x) = X^\top Q X = \langle X X^\top, Q \rangle?$$

Feasibility problem with

$$X = \begin{bmatrix} x^3 \\ x^2 \\ x \end{bmatrix}, XX^\top = \begin{bmatrix} x^6 & x^5 & x^4 \\ \cdot & x^4 & x^3 \\ \cdot & \cdot & x^2 \end{bmatrix}, A_{x^6} = \begin{bmatrix} 1 & 0 & 0 \\ \cdot & 0 & 0 \\ \cdot & \cdot & 0 \end{bmatrix}, A_{x^5} = \begin{bmatrix} 0 & 1 & 0 \\ \cdot & 0 & 0 \\ \cdot & \cdot & 0 \end{bmatrix}, \text{ etc.}$$

$b = (b_i)$ is the vector of coefficients of p.

Problems

For n-variate polynomial of degree $2d$:

$$N = \binom{n+d}{n}, \qquad m = \binom{n+2d}{n}.$$

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- interior point solvers ($\mathcal{O}(mN^{3.5} + m^2N^{2.5})$)

  CSDP, DSDP, Hypatia, MOSEK, SDPA, ...

- Alternating Direction Method of Multipliers

  CDCS, COSMO, SCS, ...

- Low rank approximation of Q, (Scaled) dominantly diagonal matrices, Newton conjugate gradient, Primal-dual hybrid gradient...

Simplifying the original program

- Block-diagonalization

  (represent Q as block diagonal direct sum of psd matrices)

  1. Chordal decomposition (exploit sparsity pattern in A_is)
  2. Wedderburn(-Artin) decomposition for matrix algebras

     (group symmetry, general *-algebras, Jordan algebras, ...)

Disclaimer: This is not new, has been re-invented several times in the literature. See e.g. Gatermann, Parrilo, *Symmetry groups, semidefinite programs, and sums of squares*, 2004

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Group symmetry invariance: Example

Robinson form

$$R(x,y) = x^6 + y^6 - x^4y^2 - y^4x^2 - x^4 - y^4 + 3x^2y^2 - x^2 - y^2 + 1.$$

R is invariant under the following operations **on monomials**

$$\alpha_1 : (x,y) \mapsto (y,x)$$

$$\alpha_2 : (x,y) \mapsto (-y,x)$$

$\{\alpha_1, \alpha_2\}$ generate a set of 8 symmetries – the dihedral group D_4.

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- symmetry of monomials $\longrightarrow$ symmetry of constraints
- symmetry of monomials $\longrightarrow$ symmetry of psd matrix Q
- invariant problem = problem "remains the same" under those symmetries

Misleading quote of the day

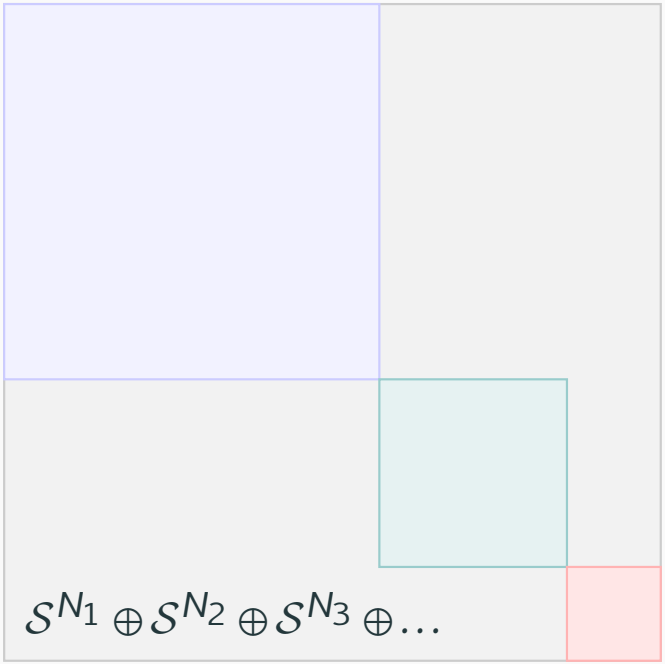
The structure of simplifications that can be derived from a symmetry pattern does not depend on the optimization problem.

Block decomposition from symmetries

Any matrix Q invariant under group symmetries admits a block-diagonal structure.

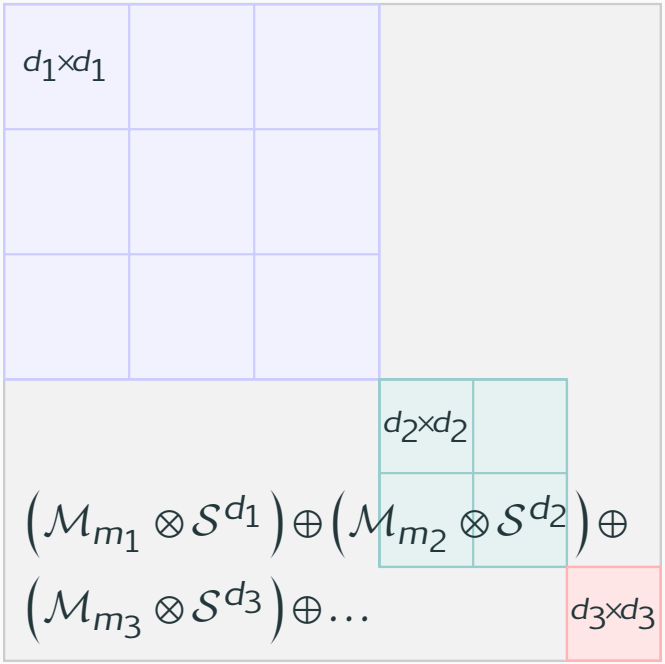
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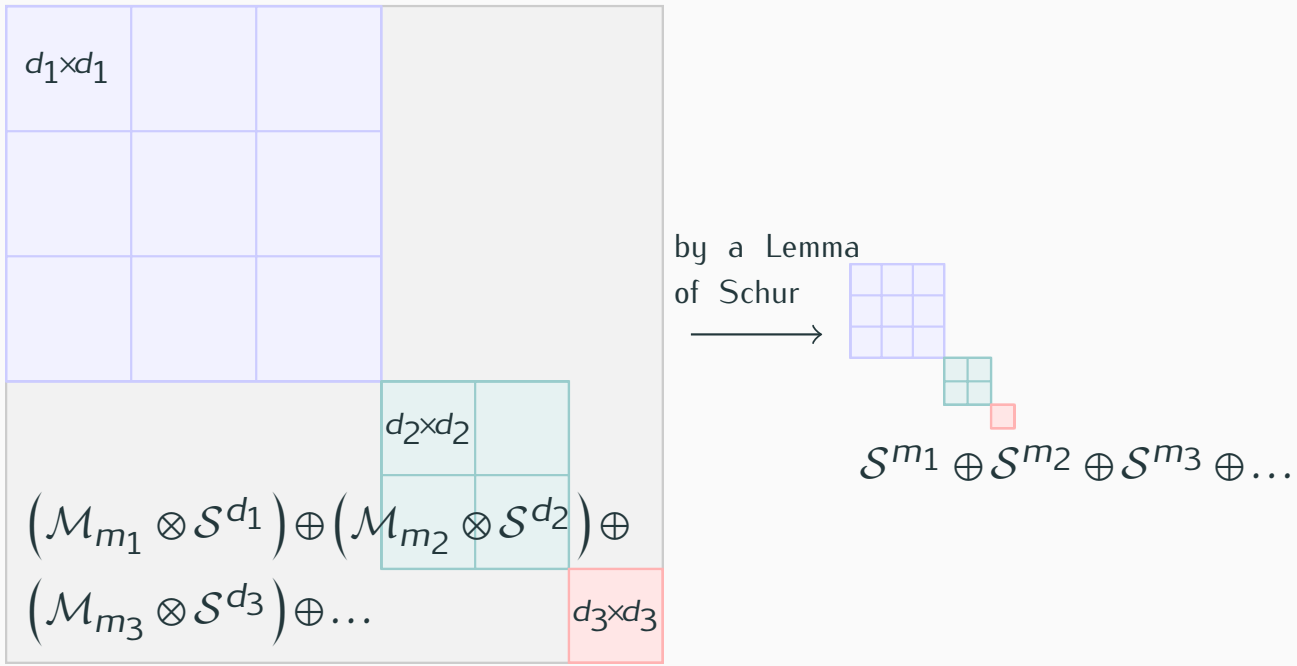
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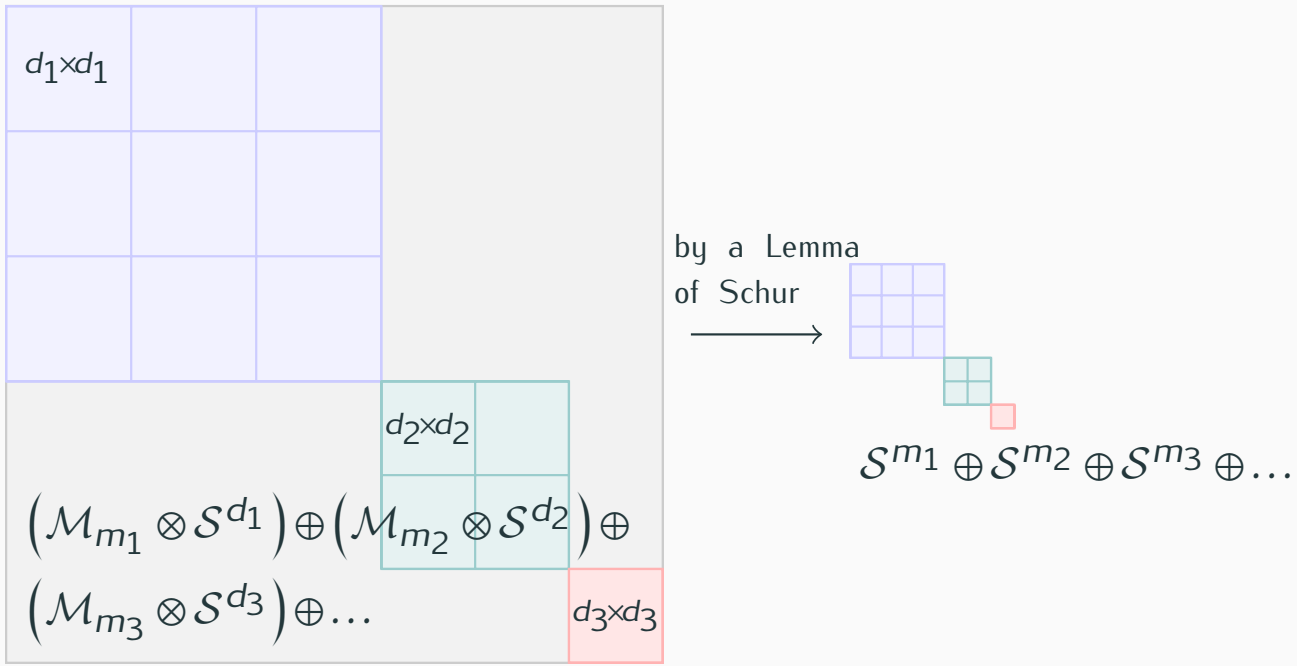
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Those isomorphims and minimal rank projections live in the group *-algebra in a basis-free form!

Example: SymbolicWedderburn.jl

```
using PermutationGroups, DynamicPolynomials
using SymbolicWedderburn
G = PermGroup([perm"(1,2)", perm"(1,2,3,4)"]) # Sym(4)
@polyvar x[1:4];  basis = monomials(x, 0:2) # 15 monomials
symmetry_adapted_basis(Rational{Int}, G, VariablePermutation(), basis,
    semisimple=true)
```

Isotypical/semisimple blocks when acting on `basis`:

$$B_1 = \begin{bmatrix} 1 \\ x_1+x_2+x_3+x_4 \\ x_1x_2+x_1x_3+x_1x_4+x_2x_3+x_2x_4+x_3x_4 \\ x_1^2+x_2^2+x_3^2+x_4^2 \end{bmatrix} \quad B_2 = \begin{bmatrix} x_1-x_4 \\ x_2-x_4 \\ x_3-x_4 \\ x_2^2-x_4^2 \\ x_3^2-x_4^2 \\ x_1^2-x_4^2 \\ x_1x_2-x_3x_4 \\ x_1x_3-x_2x_4 \\ x_1x_4-x_2x_3 \end{bmatrix} \quad B_3 = \begin{bmatrix} x_1x_2-x_1x_4-x_2x_3+x_3x_4 \\ x_1x_3-x_1x_4-x_2x_3+x_2x_4 \end{bmatrix}$$

We went from 15×15-psd constrain to sizes $(4\times 4, 9\times 9, 2\times 2)$.

Example: SymbolicWedderburn.jl

```
# [ ... ]
symmetry_adapted_basis(Rational{Int}, G, VariablePermutation(), basis
  [, semisimple=false])
```

Simple blocks when acting on `basis`:

$$B_1 = \begin{bmatrix} 1 \\ x_1+x_2+x_3+x_4 \\ x_1x_2+x_1x_3+x_1x_4+x_2x_3+x_2x_4+x_3x_4 \\ x_1^2+x_2^2+x_3^2+x_4^2 \end{bmatrix} \qquad B_2 = \begin{bmatrix} \frac{1}{3}(3x_1-x_2-x_3-x_4) \\ \frac{1}{3}(3x_1^2-x_2^2-x_3^2-x_4^2) \\ x_1x_2+x_1x_3+x_1x_4-x_2x_3-x_2x_4-x_3x_4 \end{bmatrix}$$

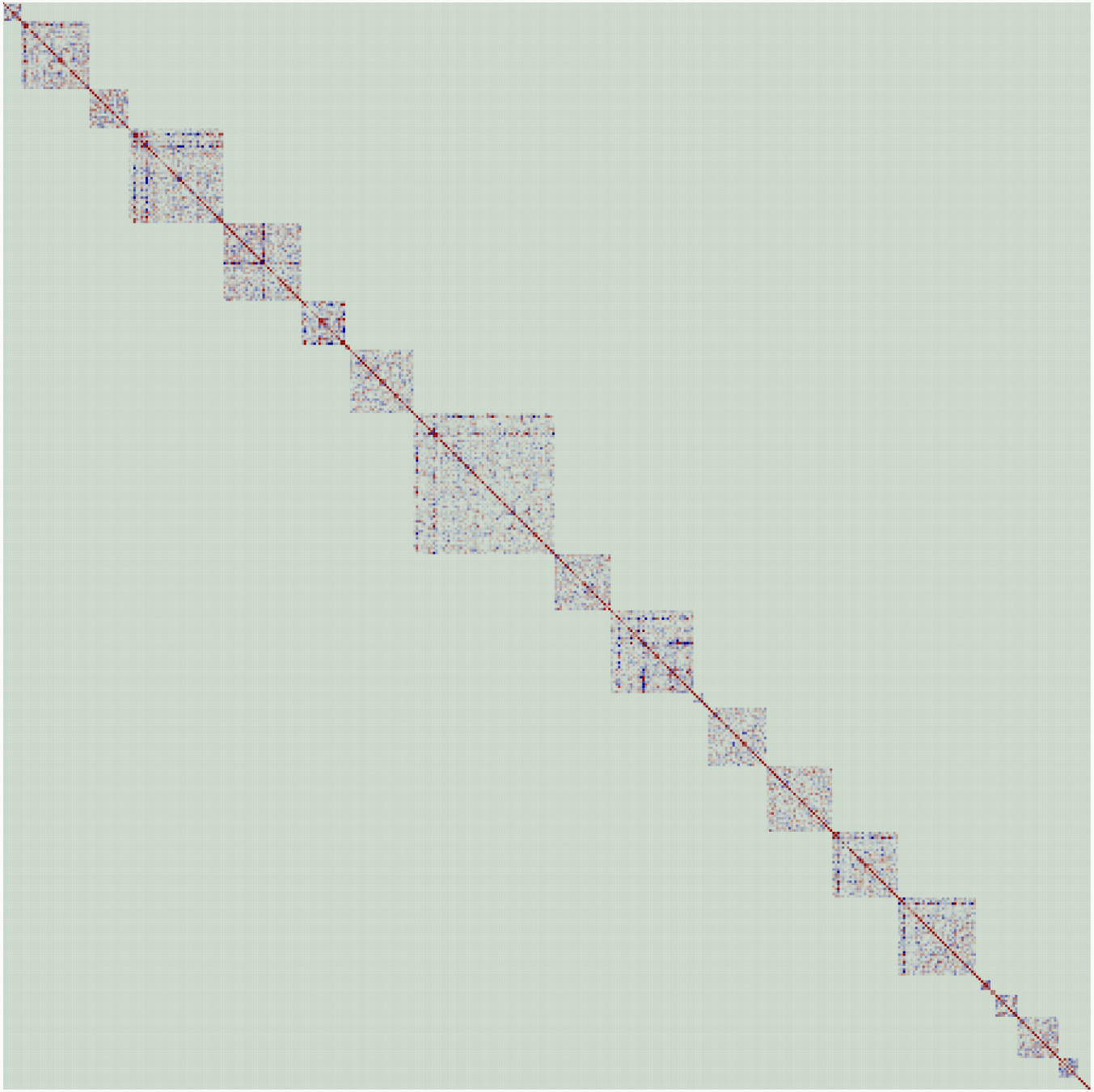
$$B_3 = \left[\tfrac{1}{2}(2x_1x_2-x_1x_3-x_1x_4-x_2x_3-x_2x_4+2x_3x_4)\right]$$

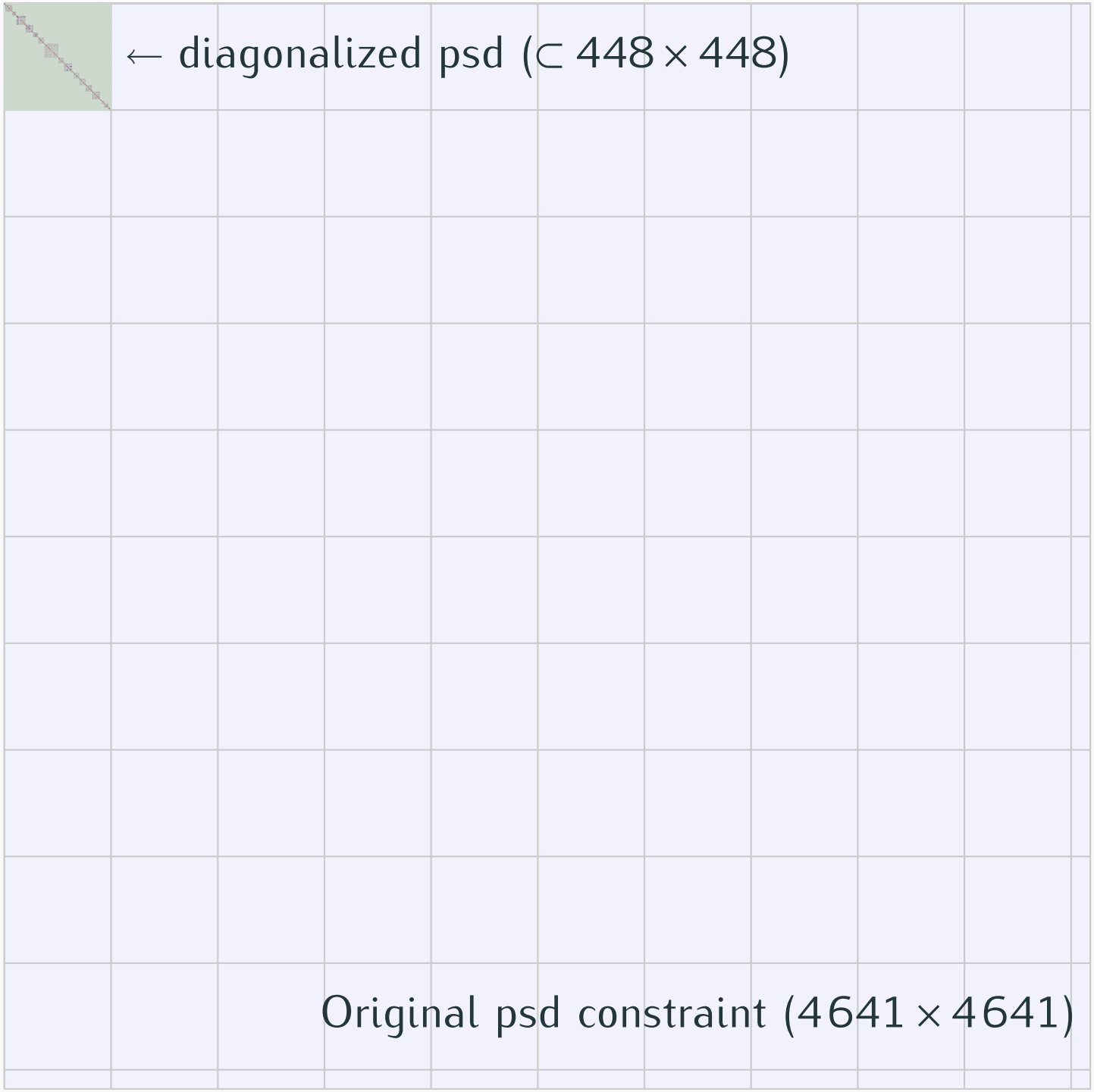
We went from $(4\times4, 9\times9, 2\times2)$-psd constraints to sizes $(4\times4, 3\times3, 1\times1)$.

Large scale example

- Optimization problem from geometric group theory[1]
- psd-constraint of size 4641×4641, $1.1 \cdot 10^7$ constraints
- symmetry group: $S_2 \wr S_5$ (3840 elements)
- After symmetrization:
  - 29-blocks (largest: 58×58), 13232 variables in total
  - 7230 constraints
- Solvable in 20 minutes to $\varepsilon \sim 10^{-12}$!

[1] Kaluba, M., Nowak, P.W. & Ozawa, N. Aut(F_5) has property (T). *Math. Ann.* **375**, 1169–1191 (2019). https://doi.org/10.1007/s00208-019-01874-9




← diagonalized psd (⊂ 448 × 448)
Original psd constraint (4641 × 4641)

```
C = ... # objective
A = ... # constraint matrices
b = ... # constraint values
basis_cnstr = ... # basis in which b is written and G acts on
basis_psd = ... # basis for the psd constraint Q

G = ... # a group, <: instance of GroupsCore.Group
action = ... # instance of SymbolicWedderburn.Action
wedderburn = WedderburnDecomposition(Float64, G, action, basis_cnstr, basis_psd

m = JuMP.Model()
psds = [ JuMP.@variable(m, [1:d, 1:d] in PSDCone())
    for d in size.(direct_summands(wedderburn), 1)]

for iv in invariant_vectors(wedderburn)
    biv = dot(b, iv)
    Ai_iv = invariant_constraint(A, iv) # ~ weighted average
    Aπs = SymbolicWedderburn.diagonalize(Ai_iv, wedderburn)
    JuMP.@constraint m sum(
        dot(Aπ, Pπ) for (Aπ, Pπ) in zip(Aπs, psds) if !iszero(Aπ)
    ) == biv
end
JuMP.@objective(m, Max, sum(dot(C, iv) for iv in invariant_vectors(wedderburn)))
```

Behind the scenes: `SymbolicWedderburn.jl`

- given action e.g. on monomials, induce the action to the whole basis
- compute character table of the group $\longrightarrow$ central projections
- for each central projection find minimal rank projection (symbolically, in the group *-algebra)
- evaluate the projections in the action on the basis $\longrightarrow$ matrix projections

Exploiting the symmetry: reducing constraints

Instead of using all $\langle A_i, Q \rangle = b_i$ it's enough to use

$$\left\langle \frac{1}{|G|} \sum_g g(A_i), Q \right\rangle = b_{[i]},$$

for each "invariant set" of the original constraints.

In the example: instead of looking at b_x (the coefficient at x), or b_y separately, we should be looking at $b_{\frac{x+y}{2}} = \left\langle \frac{1}{2}(A_x + A_y), Q \right\rangle$

This corresponds to finding a basis for the "invariants ring" $\mathbb{R}[x]^G$.